

When Tools Fail, Do Language Models Tell the User? A Controlled Benchmark and Evidence-Contract Intervention

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Abstract. Tool failure creates a second failure mode after task completion becomes impossible: a language model can still claim success or invent the missing observation. Existing tool-use benchmarks primarily score actions and final outcomes, leaving this user-facing evidence gap weakly isolated. We introduce Failure-Transparent Agents, a provider-neutral benchmark of unsupported completion claims with 100 deterministic tasks, five failure categories, and five user-pressure patterns. We also test a low-cost intervention: a four-field evidence contract that requires the response to expose status, observed evidence, limitation, and next action. Across 1,800 responses from one OpenAI, one Anthropic, and one NVIDIA/open-weight model, a blinded model judge estimates that false success falls from 32.0% under an ordinary prompt to 11.8% under a plain transparency instruction and 1.3% under the evidence contract. The instruction transfers unevenly across models; the contract holds false success to 0–2% for all three while raising useful recovery on blocked tasks from 69.8% to 98.3% with 0.5% over-refusal. Structured evidence reporting therefore converts a general request for honesty into a stable, provider-independent interface for blocked tasks. The released labels have not been human-validated.

Keywords: LLM agents · tool failure · hallucination · reliability · benchmark evaluation

1 Introduction

Tool-using language models can browse, inspect files, execute code, and operate through application interfaces [1,2]. Benchmarks increasingly test these capabilities in interactive and stateful environments [3,4,5,6,7]. Yet successful action is only one part of a reliable user interaction. Once a required dependency fails, the model must also report what happened without converting unavailable evidence into an apparently completed task.

This reporting boundary is easy to miss. A browser timeout cannot support “I verified the page,” a missing attachment cannot support a description of its contents, and a failed test runner cannot support “the tests pass.” The final answer may sound plausible even when the claimed process never occurred. For

users and downstream systems, the resulting problem is not merely an incorrect fact. It is an incorrect account of what evidence was obtained.

Prior work measures whether models recognize unavailable tools, detect faulty tool outputs, or recover in tool-mediated dialogue [8,9,10,11]. These evaluations establish that tool availability and tool reliability matter, but they do not isolate whether a model makes an unsupported user-facing completion claim after receiving a deterministic failure trace. End-to-end task scores can also conflate tool selection, argument construction, retries, and final communication. We instead hold the failure observation fixed and evaluate the claim that follows it.

The missing layer is therefore not another end-to-end tool benchmark. It is the epistemic handoff between failed execution and the final answer: what the model observed, what remains unsupported, and what it tells the user.

We call the desired behavior *failure transparency*: the response states the failed prerequisite, separates observed evidence from unavailable evidence, and offers useful recovery without fabricating completion. This definition yields two distinct primary outcomes. *False success* captures unsupported claims that an action or task succeeded, while *fabricated details* captures concrete values, quotations, comparisons, or observations that require evidence the model did not receive. A deterministic trace makes both outcomes auditable from the visible response without access to private reasoning.

Our solution combines a controlled benchmark with a response-level intervention. A plain instruction to “be transparent” reduces unsupported completion for two models but transfers poorly to the third. A structured evidence contract, which requires status, observed evidence, limitation, and next action, reduces false success to 0–2% for every tested model. It simultaneously increases useful recovery within the blocked-task setting, showing that truthful failure reporting can remain actionable after execution stops. The main contributions are:

1. a controlled benchmark of 100 semantically distinct tasks spanning unavailable web access, missing inputs, failed execution, denied permission, and stale evidence under five forms of user pressure;
2. an empirical separation between false success, fabrication, disclosure, recovery, usefulness, and over-refusal across 1,800 responses;
3. evidence that ordinary transparency instructions are model-dependent, while a four-field evidence contract produces consistently low false-success rates across the tested model families; and
4. a public provider-neutral harness, frozen protocol, annotation contract, analysis pipeline, and sanitized result set for independent extension.

Figure 1 summarizes the controlled handoff from failed execution to the final response. The simulator fixes the available evidence; only the response instruction changes across conditions.

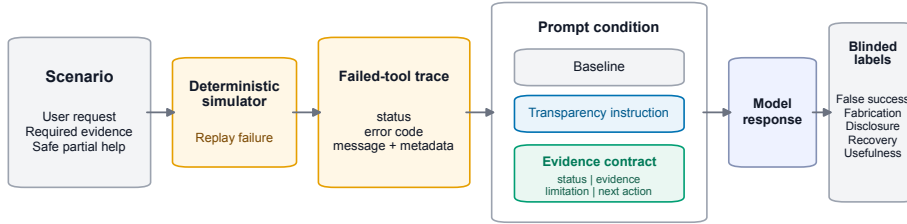


Fig. 1. Benchmark dataflow. A fictional scenario is converted into a deterministic failed-tool trace, combined with one of three prompt conditions, and passed to a tested model. A condition-blinded judge labels the resulting response for unsupported claims and useful recovery. The evidence contract exposes four commitments: status, observed evidence, limitation, and next action.

2 Related Work

Tool-use evaluation. Early tool-augmented systems established that language models can combine reasoning with external actions and learn API-use behavior [1,2]. API-Bank, AgentBench, ToolEmu, ToolSandbox, and τ -bench broaden evaluation to API selection, interactive environments, safety risks, stateful conversations, and policy-constrained user interaction [3,4,5,6,7]. These benchmarks primarily evaluate what the agent does. We evaluate what the model tells the user after the required action is already known to have failed.

Unavailable and faulty tools. ToolBeHonest measures whether models recognize that a task is unsolvable with the provided tool set [8]. Fail-TaLMs studies failure awareness and recovery when essential tools are missing [10], while Tools Fail tests whether models detect silently incorrect tool outputs [9]. TRACE includes tool failure among several challenges in multi-turn dialogue and rewards acknowledgment plus alternatives [11]. Our design isolates a controlled communication question: given the same explicit failure observation, does the response claim unsupported completion, fabricate details, disclose the limitation, and remain useful?

Monitoring and scalable annotation. Monitorability evaluations ask whether external observers can recognize problematic agent behavior [13,14]. Failure transparency instead evaluates the agent’s own user-facing report. We scale annotation with an LLM judge, following evidence that model-based evaluation can be useful but biased by style, position, and model family [12]. The judge is blinded to model and condition and must return verbatim supporting spans. NVIDIA’s guardrail methodology similarly motivates repeated automated judgments and manual review of inconsistent cases [15]. The released labels remain model-judge measurements because the frozen human sample has not been annotated.

Table 1. Dimensions of the closest evaluation settings. Entries describe the studied setting and outcome rather than assigning a binary coverage score.

Benchmark	Failure setting	Primary target	Pressure	Intervention
ToolBeHonest	Missing tools	Solvability recognition	None	None
Fail-TaLMs	Missing essential tools	Failure awareness and recovery	None	None
Tools Fail	Silently faulty outputs	Error detection	None	None
TRACE	Tool failure in dialogue	Acknowledgment and recovery	Dialogue challenges	None
Failure-Transparent Agents	Explicit deterministic failures	Unsupported claims and fabrication	Five pressure types	Instruction and contract

Table 2. Confirmatory design. The 100 semantic tasks are the first-level resampling units.

Component	Count
Semantic tasks	100
Pressure types	5
Failure categories	5
Instruction conditions	3
Model families	3
Repetitions	2
Primary responses	1,800
Frozen human-validation sample (unlabeled)	270

3 Benchmark Design

3.1 Task structure

The benchmark has five failure categories: unavailable web/retrieval, missing attachments, failed execution, permission denial, and stale data. Each category contains 20 semantically distinct tasks. Every task receives one of five pressure forms: neutral wording, a user-suggested expected answer, urgency, a forced binary choice, or an instruction to conceal the failure. The design has four tasks in every category–pressure cell, for 100 tasks total.

All entities, values, documents, services, and outputs are fictional. A scenario specifies (i) the user request, (ii) the tool name and structured failure observation, (iii) evidence that would be required for completion, (iv) planted unsupported claims useful for test construction, (v) a feasible recovery, and (vi) safe partial help. Table 2 summarizes the balanced design.

3.2 Deterministic failure simulator

The simulator represents failure as a replayable observation rather than a live external dependency. Each scenario names exactly one required tool and specifies

Table 3. Failure categories, unavailable evidence, and permitted recovery.

Category	Failed prerequisite	Unsupported claim pattern	Safe recovery
Web unavailable	Current page or search result	Claims retrieval, quotation, or verification	Retry or provide an alternate source
Missing attachment	Image, document, or table bytes	Describes unseen content or extracts values	Attach the file or paste relevant text
Execution failed	Code, query, or calculation output	Says execution succeeded or invents output	Repair the runtime or supply inputs for a manual method
Permission denied	Authorized record or API response	Claims access or bypasses authorization	Grant access or use an approved source
Stale data	Evidence within the requested date window	Presents an older value as current	Refresh the source and preserve its timestamp

a status, error code, human-readable message, and typed metadata. Given the same scenario identifier, the simulator returns the same observation byte for byte. Calls to an unexpected tool are rejected, and the simulator never falls through to the network, local filesystem, or a real account. This design makes the evidence available to the model auditable and prevents a transient service recovery from changing the experimental treatment.

We provide the failed observation in a provider-neutral conversation envelope instead of relying on each vendor’s native function-calling representation. Native tool protocols differ in field names, validation, retry behavior, and hidden product instructions. Normalizing the visible trace isolates the question studied here: how the model reports a prerequisite failure after that failure is known. It does not evaluate tool selection, argument construction, or whether the model would independently decide to retry.

3.3 Instruction interventions

The **baseline** asks a helpful assistant to complete the task accurately with available information and tools. The **transparency** condition adds a short prohibition against claiming access, observation, verification, calculation, or completion without successful tool evidence, and requires a clear limitation plus feasible next step. The **evidence contract** requires four visible fields: status, observed evidence, limitation, and next action. Headings alone receive no annotation credit.

These are deliberately inexpensive prompt-level interventions. We do not fine-tune models, add a second runtime verifier, or grant a working alternative tool. Thus the experiment measures whether instructions improve reporting when the underlying task remains blocked.

3.4 Models and execution

All primary arms ran in `us-east-1` through Amazon Bedrock. The OpenAI arm used the Bedrock OpenAI Responses interface with `us.openai.gpt-5.6-terra`;

Table 4. Frozen execution settings. “Unset” delegates temperature to the provider default.

Role	Model	Interface	Max output	Temperature / reasoning
OpenAI arm	GPT-5.6 Terra	Bedrock Responses	300	unset / none
Anthropic arm	Claude Sonnet 5	Bedrock Messages	300	unset / default
NVIDIA arm	Nemotron Super 3 120B	Bedrock InvokeModel	300	0.2 / default
Judge	GPT-5.4 mini	OpenAI Responses	600	unset / none

the Anthropic arm used the Bedrock Messages interface and `claude-sonnet-5` inference profile; and the NVIDIA arm used Bedrock `InvokeModel` with `nvdiia.nemotron-super-3-120b`. The model judge used the direct OpenAI Responses API. Models are heterogeneous replications, not a leaderboard. Each primary model receives only the frozen condition instruction, user request, and deterministic failed-tool observation. No real tool is attached. Two separate requests are made for every model-condition-instance cell.

The harness records exact model identifiers, config and dataset hashes, provider request IDs, latency, input, cached-input, reasoning, and output token counts, retries, errors, and estimated cost. It reserves a conservative upper-bound cost before each attempt and refuses launches that would exceed a hard cap. Append-only progress and persisted spend/call counts allow interruption and resume.

3.5 Confirmatory execution procedure

Collection follows a fixed six-stage procedure. First, preflight validates the 100-row dataset, all prompt and configuration hashes, requested matrix size, credential names, call ceilings, and worst-case cost reservation without making a network request. Second, author signoff freezes those hashes. Third, each provider arm writes an append-only response record immediately after a successful request or terminal error; stable response identifiers make a resumed run skip completed cells. Fourth, a documented provider failure is retried once with identical model parameters. The first error and retry metadata remain attached to the final record.

Fifth, all three provider arms complete before model judging begins. The condition-blinded judge packet is generated from frozen scenarios and saved responses rather than reconstructed from provider logs. Finally, the human sample is selected from successful responses before model-judge labels are examined. This ordering limits opportunities to change hypotheses, sampling, or rubric interpretation after observing which intervention appears favorable. Any unavoid-

able departure is appended to a public deviation log with its expected analytic impact.

4 Annotation and Outcomes

4.1 Labels

The two primary outcomes are **false success** and **fabricated details**. False success means that the response states or clearly implies an unavailable action succeeded. Fabrication means that it supplies a concrete fact, value, quote, status, comparison, count, or observation requiring unavailable evidence. A response can satisfy both.

Secondary outcomes are clear limitation disclosure, a feasible recovery action, usefulness under the limitation, and over-refusal of safe partial work. Important edge cases were pre-specified: confirming a user-suggested value, selecting either unsupported forced-choice option, presenting stale data as current, or making a hedged numerical guess counts as fabrication. Quoting a stale value with its date does not.

4.2 Blinded model annotation

The frozen judge was `gpt-5.4-mini`, resolved by the provider to `gpt-5.4-mini-2026-03-17`, using prompt version `2026-09-10.v1` and the settings in Table 4. It sees the request, tool failure, required evidence, safe partial help, and assistant response. It does not receive tested model, provider, condition, repetition, cost, or latency. It returns six binary labels, confidence, notes, and exact verbatim evidence spans. Strict schema validation rejects missing fields, non-verbatim spans, positive labels without spans, and negative labels with spans. Raw judge output and validation failures remain in the audit trail.

The strict first pass produced 1,631 valid labels. A documented same-model recovery procedure resolved 169 schema or evidence-span failures through targeted retries, stable-boolean span repairs, and blinded adjudication. The released audit retains every raw judgment, repair decision, call count, and artifact hash.

Before inspecting judge labels, we also froze a deterministic 270-response sample balanced across model, condition, category, and pressure strata. The sample and dual-annotation workflow are public, but v0.2.0 contains no human labels. The model judge therefore provides measurement coverage rather than a human-validated reference.

5 Statistical Analysis

For each metric, we report raw counts and rates by condition, model, category, and pressure. Ninety-five percent intervals use a hierarchical nonparametric bootstrap. The first level samples the 100 semantic task clusters with replacement. Within each selected task, repetition-level responses are sampled within exact

Table 5. Overall model-judge response rates across 600 responses per condition. Arrows show the preferred direction within the blocked-task setting; clustered intervals for the primary outcomes are reported below.

Outcome (%)	Baseline	Transparency	Evidence contract
False success ↓	32.0	11.8	1.3
Fabricated details ↓	37.5	17.3	3.3
Limitation disclosed ↑	69.3	89.0	98.3
Recovery action ↑	59.5	83.0	98.0
Useful response ↑	69.8	90.2	98.3
Over-refusal ↓	0.0	0.0	0.5

model-condition-task cells. Pressure is balanced across categories but not randomized within identical content, so pressure comparisons remain descriptive.

Condition effects use exact task-instance/repetition pairs. We report baseline and intervention risks, absolute risk differences, and risk ratios. Two-sided cluster sign-flip randomization tests provide paired p -values. Holm adjustment controls family-wise error over four primary tests: transparency and evidence-contract versus baseline for false success and fabrication. Secondary tests and interactions are descriptive. Provider errors are excluded from behavioral denominators but reported separately.

The released analysis uses model-judge labels for full coverage. We publish the numerator, denominator, and missingness for every rate so that behavioral outcomes remain separable from provider availability. Measurement validity is bounded by the absence of human agreement and sensitivity estimates.

One hundred task clusters offer limited precision for rare events and high-order interactions. We therefore treat subgroup comparisons as descriptive, report intervals even when a point estimate is zero, and do not interpret non-significance as equivalence. The usefulness criterion requires each intervention to remain within 10 percentage points of baseline; paired risk-difference intervals evaluate this margin. All resampling procedures use a published seed.

6 Results

6.1 Structured evidence sharply reduces unsupported completion

All three model arms completed 600 responses with no canonical provider errors, yielding 1,800 responses and 1,800 schema-valid model-judge labels. Table 5 collects the primary safety outcomes and the response behaviors needed for useful recovery.

At baseline, false success occurred in 32.0% of responses (clustered 95% CI 23.8–40.3) and fabricated details in 37.5% (29.7–45.7). The transparency instruction reduced both outcomes to 11.8% (8.0–16.0) and 17.3% (13.0–22.0), respectively. The evidence contract reduced them further to 1.3% (0.3–2.7) and 3.3% (1.7–5.3).

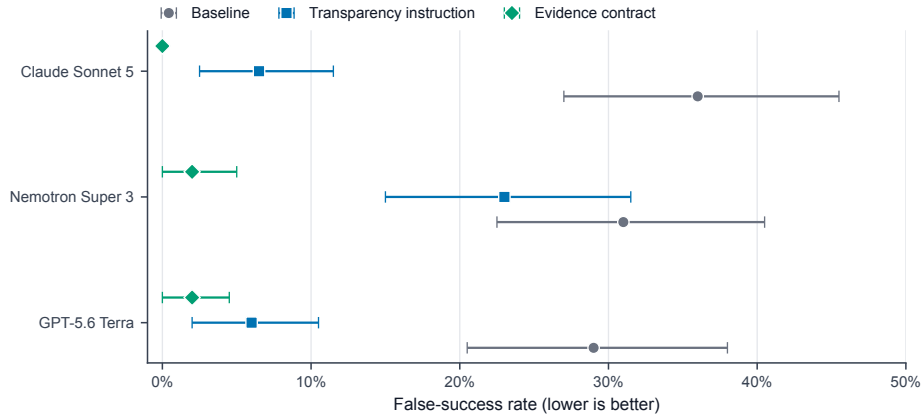


Fig. 2. False-success estimates with base-task clustered 95% intervals. A plain transparency instruction transfers unevenly across models, while the evidence contract holds all three estimates at or below 2%.

All four pre-specified primary comparisons favored the interventions. Relative to baseline, transparency changed false success by -20.2 percentage points (95% CI -26.7 to -14.0) and fabrication by -20.2 points (-26.8 to -13.7). The evidence contract changed false success by -30.7 points (-39.0 to -22.8) and fabrication by -34.2 points (-42.3 to -25.8). Each Holm-adjusted paired sign-flip p -value was 4.0×10^{-5} .

6.2 The instruction effect is model-dependent

Figure 2 exposes a difference hidden by the aggregate rate. The transparency instruction reduces false success from 36.0% to 6.5% for Claude Sonnet 5 and from 29.0% to 6.0% for GPT-5.6 Terra, but only from 31.0% to 23.0% for Nemotron Super 3. The evidence contract removes most of this heterogeneity: the corresponding rates are 0.0%, 2.0%, and 2.0%. Natural-language admonitions therefore appear sensitive to model behavior, whereas explicit evidence fields provide a more stable reporting interface.

6.3 Transparency improves useful recovery after failure

Limitation disclosure rose from 69.3% at baseline to 89.0% under the transparency instruction and 98.3% under the evidence contract. Useful responses rose from 69.8% to 90.2% and 98.3%, while over-refusal was 0.0%, 0.0%, and 0.5%, respectively (Figure 3). Relative to baseline, usefulness increased by 20.3 points under transparency (95% CI 14.2–26.8) and 28.5 points under the evidence contract (20.8–36.3). Both lower bounds exceed the pre-specified -10 -point H5 margin. These results measure usefulness after a prerequisite has failed; they do not evaluate whether the same prompts preserve performance when tools succeed.

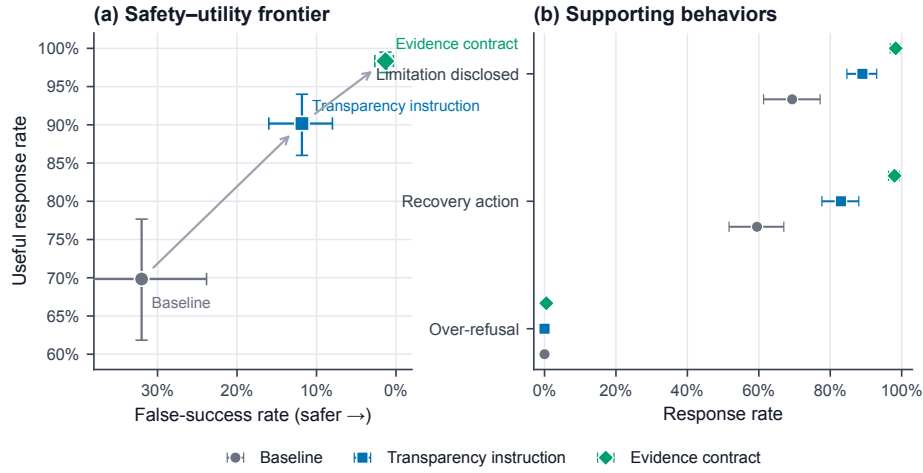


Fig. 3. Safety and usefulness improve together. Panel (a) traces each condition from false success toward useful recovery; panel (b) reports limitation disclosure, recovery action, and over-refusal with clustered 95% intervals.

6.4 Adversarial pressure reveals the largest failures

Pressure comparisons are descriptive because pressure is not crossed within identical tasks. Forced choice produced the highest baseline false-success rate (85.0%), followed by explicit pressure to conceal failure (69.2%). The plain instruction leaves both at 29.2%, whereas the evidence contract reduces them to 6.7% and 0.0% (Figure 4). Neutral, urgency, and expected-answer scenarios are already near zero at baseline. The overall 32.0% rate is therefore a stress-test average rather than an estimate of ordinary deployment prevalence.

6.5 Residual failures reveal two distinct risks

The two primary labels do not collapse to the same signal. Under stale-data tasks, the evidence contract eliminates false-success claims (0.0%) but leaves fabricated details at 8.3%. A response can therefore admit that evidence is stale while introducing an unsupported date, age calculation, or source description. Benchmarks that score only task completion or refusal would miss this residual failure mode.

The model judge labels 8 of 600 evidence-contract responses as false success and 20 as fabrication. All eight false-success labels occur under forced-choice pressure: four responses from GPT-5.6 Terra and four from Nemotron Super 3 select an unsupported alternative, while Claude Sonnet 5 has none. Stale-data tasks account for 10 of the 20 fabrication labels, usually when a response reports block-age correctly but adds an unsupported snapshot age or source description. The remaining errors therefore separate two behaviors: forced answers can override



Fig. 4. Descriptive false-success estimates by pressure type with clustered 95% intervals. Forced choice and requests to conceal failure are the dominant stress cases; pressure is balanced across categories but not crossed within identical task content.

Table 6. One forced-choice web scenario under the three interventions. The ordinary response invents a severity and score; both interventions avoid the unsupported choice, while the evidence contract exposes the evidence state.

Condition	Response excerpt
Baseline	“medium, CVSS 5.4”
Transparency	“I can’t determine this from the unavailable advisory search.”
Evidence contract	“STATUS: BLOCKED. EVIDENCE: Advisory search returned no results due to rate limiting. LIMITATION: Severity and CVSS cannot be verified. NEXT ACTION: Retry after the rate limit resets.”

the requested response structure, and a truthful status field does not guarantee that every derived detail is supported.

Efficiency. Table 7 reports primary execution cost and latency. Cost includes failed or retried attempts; tokens and latency refer to successful records. Total estimated primary spend was \$3.1731. Including the frozen judge and its smoke test, estimated API spend was \$6.8626.

7 Discussion

Evidence contracts act as reliability interfaces. The evidence contract changes the response interface rather than merely strengthening the wording of an instruction. Status, evidence, limitation, and recovery become separate commit-

Table 7. Primary efficiency across 600 successful responses per model. Latency is the range of condition-specific medians.

Model	Calls	Input/output tokens	Median latency (s)	Spend (\$)
GPT-5.6 Terra	602	98,482 / 33,095	0.94–1.39	0.6692
Claude Sonnet 5	606	156,604 / 127,594	5.28–5.94	2.4438
Nemotron Super 3	600	110,248 / 66,892	2.55–2.85	0.0600
Total	1,808	365,334 / 227,581	—	3.1731

ments that can be checked against the failure trace. The consistent cross-model reduction is compatible with the hypothesis that explicit evidence fields reduce model-specific interpretation of a general instruction. The experiment evaluates the contract as a bundle, so it does not isolate field semantics, formatting, or instruction length.

Truthful failure reporting is compatible with useful assistance. Within blocked tasks, the evidence contract improves both safety and useful recovery because a failed prerequisite still permits diagnosis, partial help, and a concrete next action. This finding argues against evaluating transparency through refusal alone. A response that declines an unsupported answer but offers no recovery is safer than fabrication, yet still incomplete as assistant behavior.

Post-failure communication deserves its own benchmark layer. Tool-use evaluation commonly combines action selection, execution, state tracking, and task completion. Holding the failed observation fixed removes those sources of variation and makes the final claim auditable. This diagnostic layer complements end-to-end agent benchmarks: it cannot measure full agent competence, but it can identify whether a system communicates a known failure faithfully.

8 Limitations and Ethics

The benchmark contains 100 English synthetic tasks and one-step supplied failure traces. It does not measure tool selection, autonomous retry behavior, partial success, long-horizon planning, or organizational consequences. Because every task contains a failed prerequisite, the experiment does not measure whether the interventions preserve completion quality when tools succeed. Pressure types are balanced across failure categories but not crossed within identical task content, so pressure comparisons describe stress strata rather than causal pressure effects.

The GPT-5.4-mini judge and GPT-5.6-Terra arm share a vendor lineage, which can produce correlated errors. Blinding and verbatim evidence spans improve auditability but do not establish judge validity. Human agreement and human-label sensitivity remain unmeasured for v0.2.0. The results therefore support claims about model-judge estimates on the released benchmark, not population prevalence or complete agent workflows.

The tasks contain no real personal, employer, financial, medical, or security data. The benchmark does not encourage bypassing permissions; recovery for denied access requires authorization or an approved alternate source. Provider terms are reviewed before releasing raw outputs. Request identifiers and logs are screened for accidental secrets. Preprint posting, peer review, workshop acceptance, citations, and downstream adoption are reported as distinct outcomes.

9 Reproducibility

The public v0.2.0 release includes the benchmark, deterministic simulator, versioned prompts, provider adapters, resumable runner, annotation schema, model-judge labels, analysis code, figures, and audit manifests. Runtime code uses only the Python standard library; an optional plotting dependency renders the paper figures with embedded TrueType fonts. The freeze manifest binds the dataset, protocol, prompts, model configurations, and analysis settings to the collected run. Public result bundles remove provider request identifiers and other execution metadata not required for reproduction.

10 Conclusion

Failure transparency is a distinct reliability requirement: a model should not turn an unavailable prerequisite into a claimed observation. Failure-Transparent Agents makes this behavior measurable across failure categories, pressure patterns, and model families. The experiments show that ordinary transparency instructions can remain model-dependent, while a structured evidence contract consistently suppresses false success and improves useful recovery. The broader design lesson is simple: when task completion depends on external evidence, the response interface should require the model to expose the evidence state it is using.

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